

Institute of Actuaries of India

Subject CP3 – Communication Practice

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

Dear Sir/ Mr.

Thank you for your mail. I have attached a note as requested by you, explaining our approach to reinsurance. Please do get in touch with me if you need any further clarifications.

Regards,

XXX

Note on reinsurance arrangements**1. Need for reinsurance****Managing our profits and capital**

We use estimates for death rates or hospitalisation or sickness rates depending on the type of product, in pricing and setting up of liabilities on our books. In arriving at these estimates we use our past experience and adjust it for any known or anticipated changes in the underlying rates. If we receive actual claims as per these estimates, then our profits will also be in line with our expectations. However, we cannot rule out the possibility of some variation in the claims which we receive. This could be because the underlying rate which we estimated itself is not appropriate or there could be some random variability year on year. Thus, profits could vary on a year-on-year basis. In addition, due to the underlying sums assured in our term products varying from relatively small amounts to ₹100 crores exposes us to the risk of claims arising from large cases in some years. Therefore, even if the underlying rates are correct if we retain all the business with us, we would be exposed to this variability on claims and our profits. With reinsurance the variability of profits can be reduced and profits can be more stable year on year. The capital requirement is also reduced due to reduction of volatility of claims.

Technical assistance

We have also entered into newer areas of critical illness cover and hospitalisation covers. We do not have any experience with these kinds of products and therefore we are dependent on the reinsurers for their expertise. We need technical support for design of the contracts, pricing of the contracts and even underwriting, that is the screening which we do before we onboard the customers.

2. Level of reinsurance

Our Board risk appetite statement limits the level of exposure to risks such that in case of an adverse impact our earnings should not be impacted by more than 10% from the base level.

The key risk in our products is the mortality risk. Reinsurance reduces our overall level of profitability if our claims are in line with our expectations as reinsurance premiums include costs of reinsurers and their profit margins. However as explained it limits our losses in case our estimates turn out to be wrong for whatever reason. We have considerable experience in this book which we can rely on and therefore we can retain more of the risk on our books. The key is therefore to find an optimum level of reinsurance which balances our need for stable profits while reducing the costs of reinsurance. To understand the impact on variability of profits we have conducted some sensitivities at different levels of reinsurance support and then checked the impact on profits if the claim experience varied from the expected levels. The level of variability that we need to check is also as per the risk appetite statement of the Board. Based on the results of this analysis we have recommended a retention of ₹ 50 lakhs as that level of reinsurance

is optimum as compared to the level of fluctuations in our profits in case of an adverse experience. In our unit linked products we only take reinsurance for the portion in excess of the unit fund we which would need to pay in case of death. This also helps us reduce our reinsurance cost.

3. Types of arrangement

For the death cover based on the analysis above we have an arrangement whereby any cover in excess of ₹ 50 lakhs is passed on to the reinsurer.

For the relatively newer products we have opted for a “quota” share arrangement with the reinsurer. This arrangement lets the Company share the risk with the reinsurer in a proportionate way. However, for the critical illness product, the sums assured can be as high as ₹ 1 crore though we still have to see the level of covers which would be opted for by the customers. For this product while we will share the risk with the reinsurer proportionately, we are also limiting the maximum exposure to the company to ₹ 10 lakhs with the balance being passed on the reinsurer. For our hospitalization cover since the maximum reimbursement or sum assured is capped to ₹ 10 lakhs, we are comfortable with a proportionate cover. As and when we have some experience in these products, we would be evaluating the risks which we retain.

In addition, for our death cover we have a catastrophe reinsurance cover. As our business increases, and we take on more customers on our books, we are exposed to the possibility of a high number of claims in case of a catastrophe. While the portion which is reinsured would be paid by the reinsurer in case of catastrophe, the risk arises from the portion which we are retaining. Therefore, in order to protect ourselves from these rare events we have a catastrophe cover of ₹ 100 crores. In the unfortunate event of a catastrophe we would be reimbursed from the reinsurance company up to this limit on our claims due to the catastrophe.

4. Other qualitative aspects

We enter into reinsurance arrangements with reinsurance companies which have credit ratings of at least AA-. We also ensure that if we guarantee premiums to the customers for a period of time the reinsurer also does the same. .

[90 Marks]

Solution 2: Any of the terms as identified by the candidate with suitable explanations. Some of the terms may not be exactly jargon however, in the context of finalizing the reinsurance program may need detailed explanations.

e.g.

Risk appetite as defined in the context of the example given here

Credible experience

Parameter risk

Volatility

Stress scenarios

Deterministic approach

Retention levels,

Types of reinsurance arrangements

1 in 10 scenario

[5 Marks]

Solution 3: Examples with explanation of any information that the candidate considered irrelevant

It was really not necessary for the Director to understand how the levels had been determined. The Director's concerns was that reinsurance would result in reduced profits. Explaining that this is one of the factors considered and summarizing the results of the sensitivity analysis is sufficient to address the Director's question.

[5 Marks]
